

Yunzhi Lin

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EDUCATION

University of California, Berkeley

B.S. Bioengineering

GPA: 3.44/4.00

Expected Graduation: May 2028

Coursework: Electronics for IoT, Mechanical Behavior of Materials, Polymeric Materials, Properties of Materials, Biomechanics, Computational Biology, Biochemistry, Tissue Engineering, Structure and Interpretation of Computer Programs

SKILLS

Software: SolidWorks, CREO, Windchill, Fusion 360, OnShape, 3DEXperience, FEBio, Ansys, Python, Java, SQL, Git, Scheme

Techniques: DFMA, GD&T, 3D Printing, Bioprinting, Lean Manufacturing, Machine Maintenance and Improvement, Good Manufacturing Practice (GMP), Good Documentation Practice (GDP)

EXPERIENCE

Tesla

Fremont, CA

Process Engineering Intern - Optimus Actuators

08/2026 – 01/2027

- Driving Optimus actuator production ramp-up by analyzing process data with SQL and visualization tools, performing root-cause analysis, designing fixtures, developing manufacturing instructions, and collaborating cross-functionally to improve yield.

GE HealthCare

Waukesha, WI

Mechanical Engineering Intern - Maternal Infant Care

01/2026 – 07/2026

- Designed 13 production-ready GD&T drawings for sheet metal and injection-molded parts in Creo, performed tolerance stack-up analysis, Ansys FEA, and DFMA reviews to support NPI electromechanical integration and manufacturing readiness.
- Assessed suspended-mass safety risks across 90+ components using energy calculations and FMEA, implementing secondary retention, fail-safe mechanisms, and poka-yoke design features to improve compliance with medical device safety standards.
- Designed test fixtures, developed and executed DOEs, and analyzed data for thermal management, center-of-gravity, sensor, and acoustic testing to support design verification and validation in compliance with ISO, IEC 60601, and system requirements.
- Co-published a defensive publication proposing an AI-integrated suspended-mass analysis tool to improve workflow efficiency.

Silver Lake Research Corporation

Irwindale, CA

Engineering Associate

09/2024 – 08/2025

- Authored and updated comprehensive SOPs for operation, maintenance, and calibration of 5 manufacturing machines, standardizing procedures to align with GMP and improve training, reliability, and audit readiness.
- Developed a scheduling system to centralize task assignments across Manufacturing, Production, and QC, improving real-time task visibility, reducing scheduling conflicts, and accelerating project handoffs.
- Reduced end-to-end production lead time by 18% and minimized material waste by leading root-cause analyses on recurring defects and executing targeted equipment and process modifications.
- Designed custom machine fixtures in OnShape, prototyped them with 3D printing, and applied GD&T to improve machine usability and production repeatability.

Engineering Intern

05/2024 – 08/2024

- Authored and executed 7 validation protocols for new manufacturing machines and software tools, completing 7 verification and validation reports to confirm system performance under varying constraints and enhance QC efficiency.
- Developed a cost analysis and production capacity evaluation tool in Excel for 4 products, integrating Macros and VBA Scripts to allow testing of different constraints and processes, resulting in more cost-efficient production workflows.
- Designed a 3D production layout for assemblies using Creo, optimizing space utilization, material flow, and environmental control, leading to increased operational efficiency.
- Led maintenance and troubleshooting on key manufacturing machines, enhancing reliability and reducing downtime while developing technical expertise in equipment management.

UC Berkeley Mechanical Engineering

Berkeley, CA

Undergraduate Research Assistant — advised by Professor Grace Gu

02/2025 – 12/2025

- Fabricated collagen-alginate 3D-bioprinted composite scaffolds across multiple blend ratios to emulate ligament mechanics and evaluated material properties through tensile and compression testing.
- Optimized printing and post-processing parameters to improve strength, shape stability, and durability of scaffold prototypes.
- Tuning collagen and collagen-alginate mechanics through extrusion bioprinting process parameters, RSC Advances, 2025.

PROJECTS

IoT Electromechanical Marble Maze

10/2025 – 12/2025

- Developed a networked mechatronic maze with PID-controlled 2-axis tilt, 4 randomized PWM servo gates, and sensor-driven game logic, integrating ESP32 MicroPython firmware and validating closed-loop behavior through iterative testing and tuning.
- Designed and fabricated the mechanical system using Fusion 360, 3D printing, and laser cutting, and implemented ESP-NOW wireless communication and HTTP-based game data logging to a Google Apps Script-hosted live leaderboard.